BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 027 · IP and MAC Addresses

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. How does wired Ethernet differ from Wi-Fi?

Q2. What does a server do in a client-server network?
A. Only requests services
B. Replaces every client
C. Provides services to clients
D. Only connects cables

Q3. Add 00010111 and 00000110 as unsigned binary numbers. Give the eight-bit result.

Q4. (a) Logically shift unsigned eight-bit 00110001 one place left. Give the eight-bit result and denary value. (b) Logically shift unsigned eight-bit 00110001 one place right. Give the eight-bit result and denary value.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

IP address — Network address used to identify and route to a destination.

IPv4 — A 32-bit IP addressing format typically written as four decimal octets.

IPv6 — A larger IP addressing format written using hexadecimal groups.

MAC address — An address identifying a network interface, typically six hexadecimal pairs.

Switch — Local network device using MAC addresses for forwarding.

Router — Device using IP addresses to route between networks.

New learning

IP addresses: An IP address identifies a network destination and allows routers to forward traffic between networks. IPv4 is commonly shown as four decimal numbers.

MAC addresses: A MAC address identifies a network interface; it is commonly shown as six pairs of hexadecimal digits.

Switch and router jobs: Within a LAN, switches forward based on MAC addresses. Between networks, routers use IP addresses.

Compare examples: IPv4 and MAC addresses have different roles and formats. Do not confuse a device's IP address with the MAC address of its NIC.

Worked example

A computer sends data to a website on another network. Explain why both MAC and IP addressing may be used. [4 marks]

1. The computer first communicates on its local network, then needs to reach a remote network.

2. Local network interfaces use MAC addresses [1], allowing switching towards the local gateway [1].

3. Routers use IP addresses [1] to forward data between networks towards the destination [1].

4. Match four points to local MAC use and remote IP routing.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Q3. IP addresses identify network destinations [1] for routing [1], and routers use them to forward between networks [1]. MAC addresses identify local interfaces [1], and switches use them [1] for local forwarding [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Ethernet uses cables. Wi-Fi uses radio, allowing movement but risking interference.

Q2. C — Provides services to clients

Q3. 00011101; 23 + 6 = 29.

Q4. (a) 01100010 = 98. Fill the empty bit with 0. (b) 00011000 = 24. Fill with 0 and lose the old rightmost bit.

I Do / We Do pages · Guided practice

Q1. A network interface has a MAC address [1] identifying it locally [1]. A switch uses MAC addresses [1] to direct traffic to the intended LAN port/device [1].

Q2. Destination IP address identifies the remote network destination [1]. Routers inspect IP addressing [1] and forward packets between networks [1] towards the correct server [1].

Q3. 10.2.4.8 is IPv4 [1] shown as four decimal values [1]. 5A:01:2B:0C:34:EF is a MAC address [1], shown as six hexadecimal pairs [1].

You Do pages · Code First

```
octet = 260
if octet > 255 then
    print("Outside IPv4 range")
endif
```

Expected result: Outside IPv4 range

You Do pages · Independent questions

Q1. Identifies a network destination/used for routing [1].

Q2. The destination computer's NIC has a MAC address [1], which identifies it on the LAN [1]. The switch uses the destination MAC address [1] to forward the data to the appropriate port towards reception [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

A printer and PC share one LAN. Explain why MAC addresses are useful to a switch. [4 marks]

Worked method: A network interface has a MAC address [1] identifying it locally [1]. A switch uses MAC addresses [1] to direct traffic to the intended LAN port/device [1].

Guided 2

A college connects its LAN to an external website. Explain how IP addresses support the connection. [4 marks]

Worked method: Destination IP address identifies the remote network destination [1]. Routers inspect IP addressing [1] and forward packets between networks [1] towards the correct server [1].

Guided 3

A technician sees 10.2.4.8 and 5A:01:2B:0C:34:EF. Compare these two addresses. [4 marks]

Worked method: 10.2.4.8 is IPv4 [1] shown as four decimal values [1]. 5A:01:2B:0C:34:EF is a MAC address [1], shown as six hexadecimal pairs [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Predict the message printed.

```
octet = 260
if octet > 255 then
    print("Outside IPv4 range")
endif
```

predict

Question 1

State one role of an IP address. [1 mark] Source: Adapted from your Paper 1 revision booklet, 1.3.2.

Question 2

A museum ticket scanner sends entry data to the reception computer on the same LAN. Explain how MAC addresses help the switch forward this data. [4 marks] Source: Original J277-style question, not an OCR past-paper original.

Question 3

Compare IP addressing and MAC addressing and explain their use by routers and switches. [6 marks] Source: Original J277-style question, not an OCR past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

A router primarily uses destination IP addresses to forward between networks.